

A level economics

chp:- Demand and Supply

Topic:- PED and Total revenue

alpha

Zarak
Mushtaq



Price Elasticity of demand and Total revenue

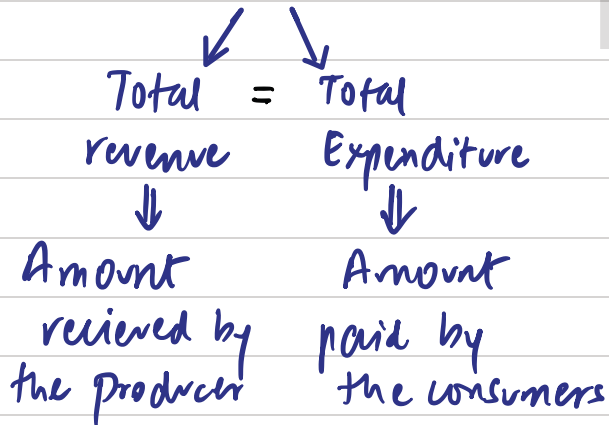
* Difference between profit, sales and revenue

Total revenue

↓
Total income of the producer

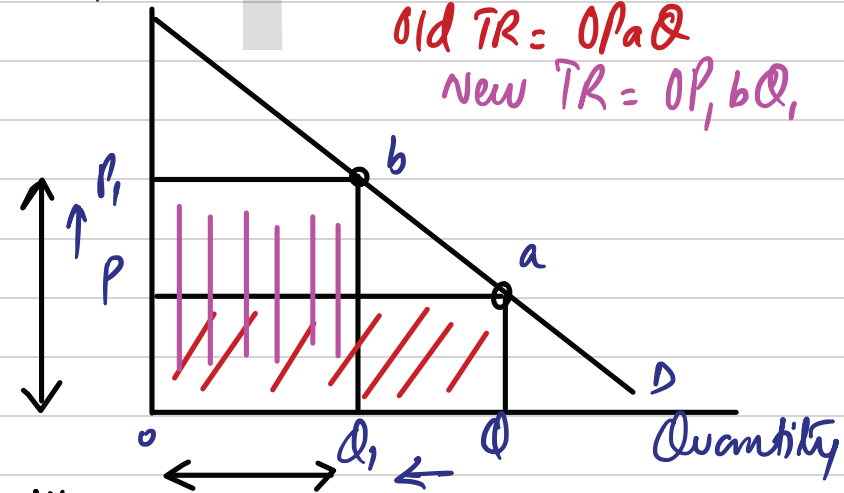
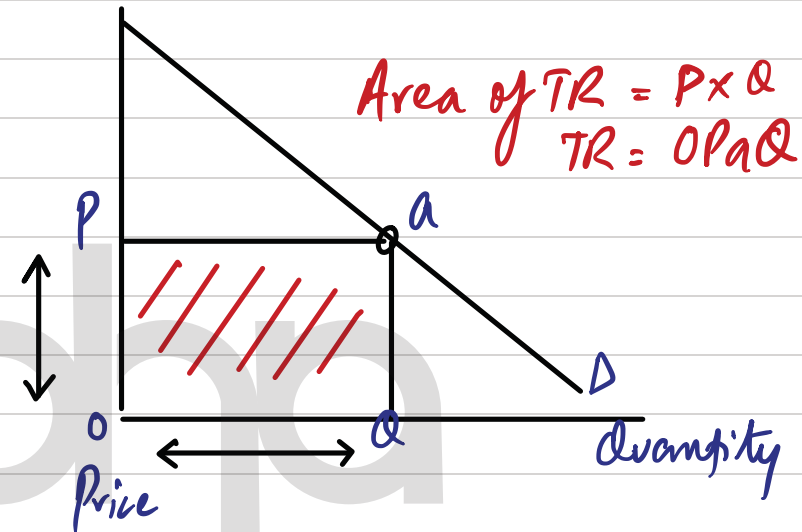
$$TR = \text{Price} \times \text{Quantity} \rightarrow \text{sales}$$

$$\$5 \times 100 \Rightarrow \$500$$

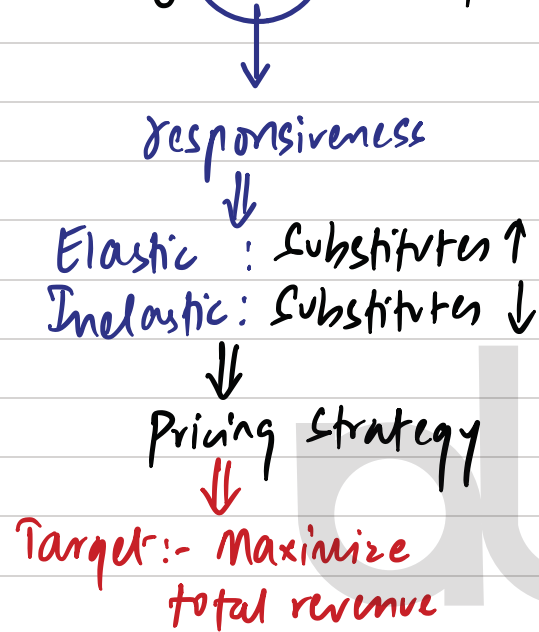


Note:- Total revenue is equal to total Expenditure

Price



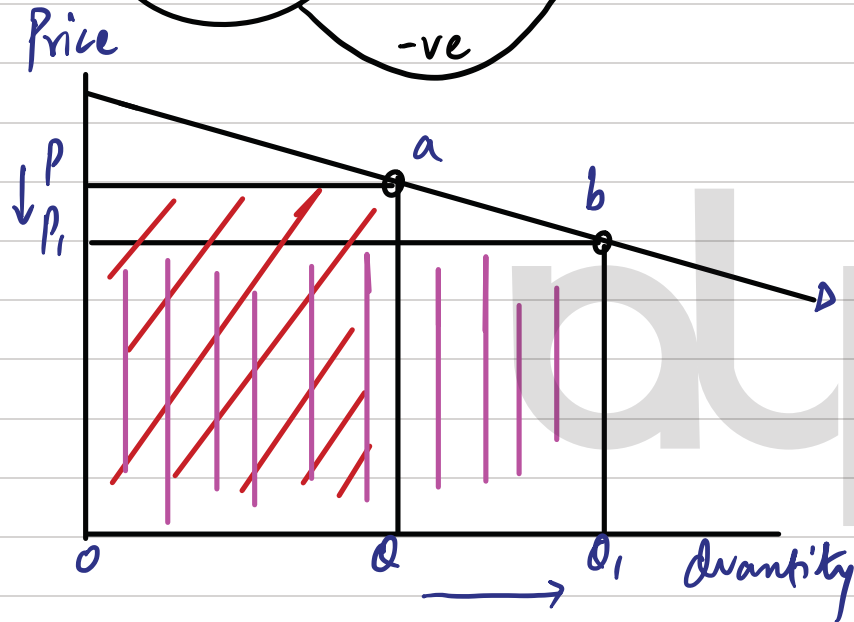
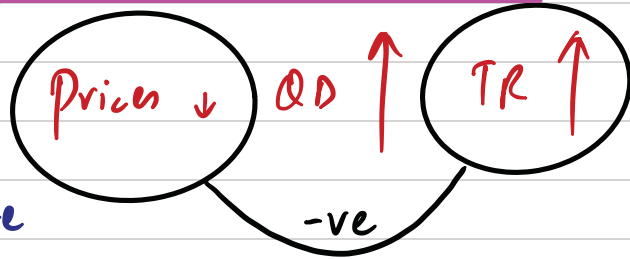
Usefulness of PED to a producer



Note:- A producer will always aim to maximize its total revenue, for which it will utilize the information of price elasticity of demand, and plan separate pricing strategy to maximize total revenue.

Note:- Price discrimination, charging different prices to different customers for the same product, reasons not associated with cost, it depends on price elasticity of demand.

If demand is elastic :-



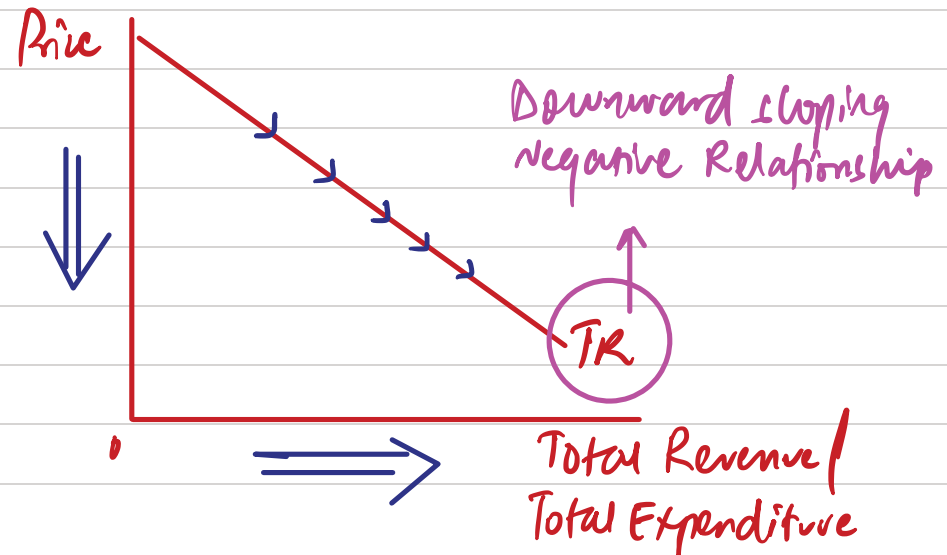
Old TR :- $OPaQ$

New TR :- OP_1bQ_1

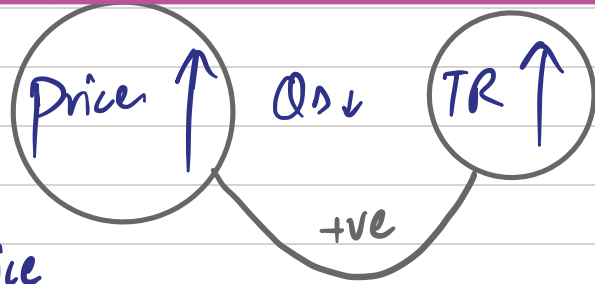
TR ↑

Note:- In case of elastic demand, where a small change in price brings a larger percentage change in quantity demanded because of availability of substitutes, a producer will reduce the price by smaller percentage which will extend quantity demanded by greater percentage, as shown in the diagram.

Note:- There is a negative relationship between price and total revenue (total expenditure) in case of elastic demand.



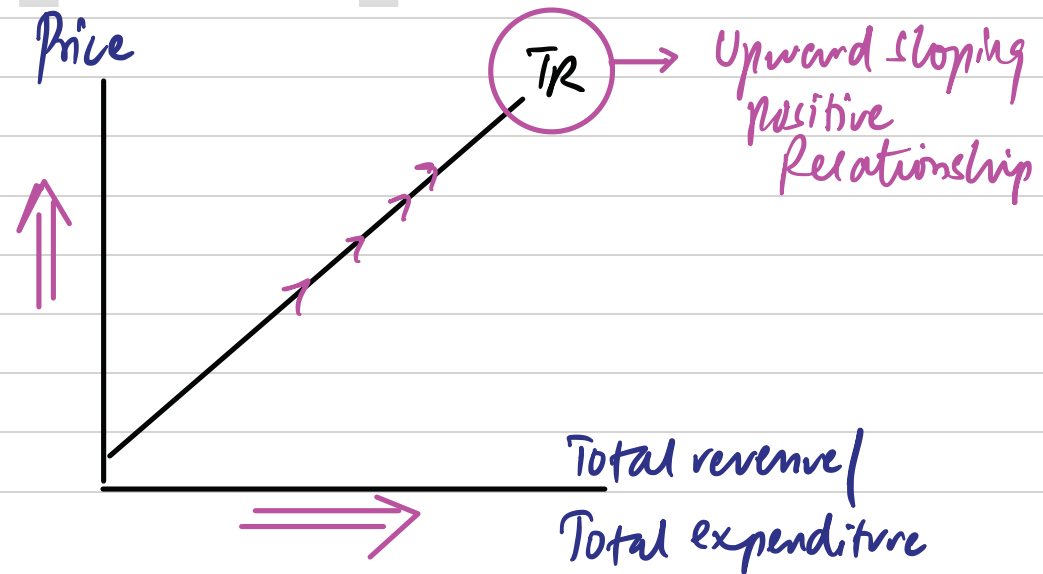
If demand is inelastic :-

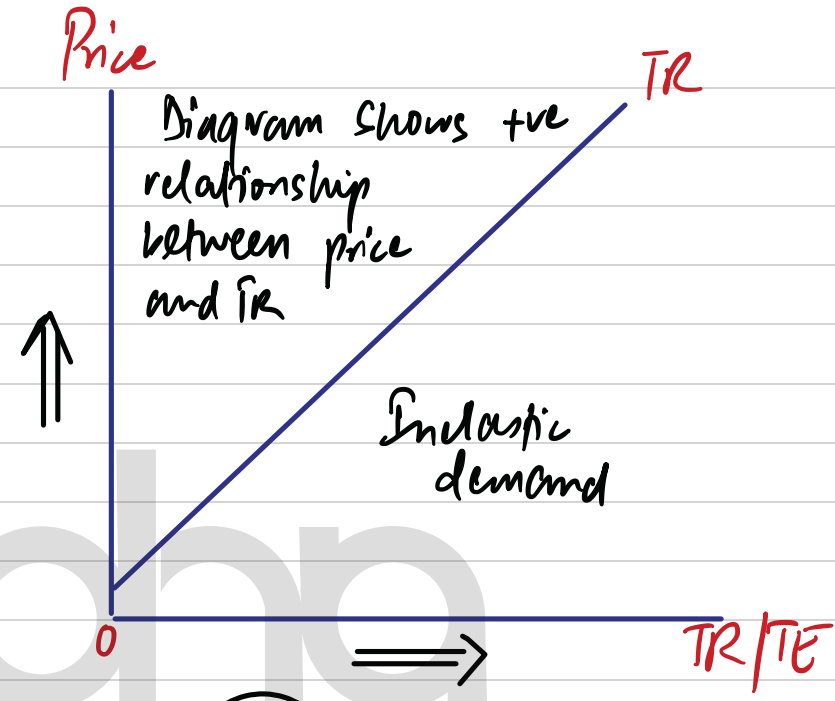
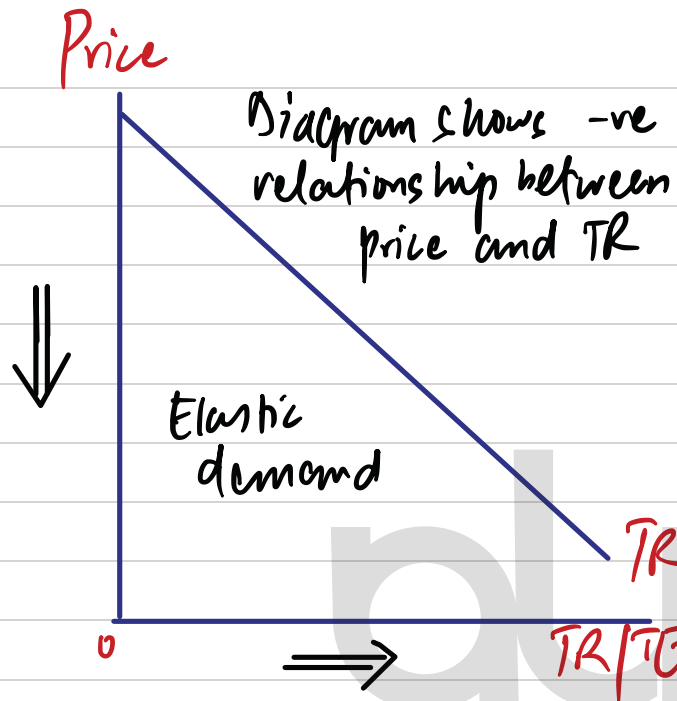


Old TR :- $OPaQ$
 New TR :- OP_1bQ_1
 TR ↑

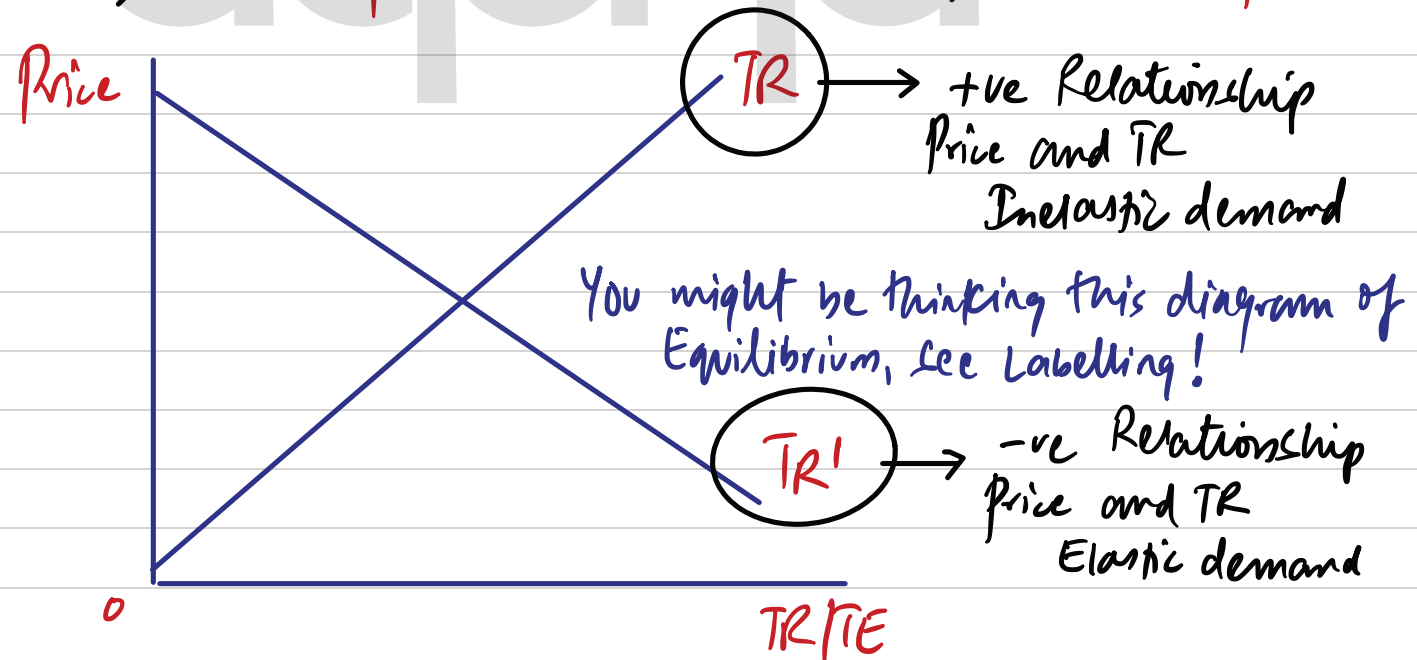
Note:- In case of inelastic demand, where a larger change in price brings a smaller percentage change in quantity demanded, due to availability of less substitutes, a producer will increase the price by greater percentage which will contract quantity by smaller percentage, causing a rise in total revenue.

Note:- There is a positive relationship between price and total revenue (total expenditure) in case of inelastic demand.





* See the labelling before judging a diagram



Important

Elastic :- price $\downarrow$ $QD \uparrow$ $TR \uparrow$

Relationship, price and QD :- Negative (Downward sloping graph)

price and TR :- Negative

Qty and TR :- positive (upward sloping graph)

Inelastic :- price $\uparrow$ $QD \downarrow$ $TR \uparrow$

Relationship, price and QD :- Negative (Downward sloping graph)

price and TR :- positive (upward sloping graph)

Qty and TR :- Negative (Downward sloping graph)

Questions (H.w #5)

Marks:- 26

- Q1 Define total revenue ? 2 → Define
- Q2 Define price discrimination ? 2 → Define
- Q3 Explain the aim of a producer ? 2 → paragraph given
- Q4 Explain how a producer can maximize its total revenue if demand for the product is elastic ? 6 Define elastic, strategy, 2 diagrams ($P \downarrow Q \uparrow TR \uparrow$)
- Q5 Explain how a producer can maximize its total revenue if demand for the product is inelastic ? 6 Define inelastic, strategy, 2 diagrams ($P \uparrow Q \downarrow TR \uparrow$)
- Q6 (next page)



Q6 Using elasticity of demand, explain why a train company might introduce a policy of raising fares at 'less busy' travel times and lowering fares at less busy travel times? &

Ans format:- Define PED, formula, Define elastic and inelastic (Paragraphs given)
 Aim of the producer (Paragraph given)
 Strategy for elastic demand, 2 diagrams [Total 3 diagrams]
 Strategy for inelastic demand 2 diagrams
 * Application marks:- Connect to train company!

